TG-051

Cavitation and NPSH Troubleshooting Guide

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1 Purpose

This guide supports diagnosis of cavitation and net positive suction head deficiencies in centrifugal pumps. It is diagnostic only; corrective work is governed by MM-207 and operational response by the applicable operating procedure.

2 Recognising Cavitation

2.1 Acoustic and Vibration Signature

Classic cavitation presents as a gravel-like noise at the pump casing accompanied by broadband vibration energy between 2 and 20 kHz. Absence of audible noise does not exclude cavitation, particularly where the condition is intermittent.

2.2 Continuous versus Intermittent

Continuous cavitation produces a stable degraded head and steady noise. Intermittent cavitation produces a repeating alarm pattern with normal operation between events, and is frequently misattributed to instrument fault or to a primary bearing defect.

2.4 Alarm Pattern Signature

Repeating low suction pressure events accompanied by rising bearing temperature within 10 to 20 minutes are characteristic of intermittent cavitation. Available NPSH should be recalculated against the pump curve at current duty before any mechanical intervention is scheduled.

The ordering of the two signals is diagnostic. Where the suction event consistently leads the bearing event, the bearing signal is a consequence and the suction side is the correct target for investigation. Where the bearing event leads, suspect a primary bearing or lubrication fault and refer to MM-207 §6.

Observed pattern	Likely primary cause	First action
Suction leads bearing by 5 to 30 min	Suction-side restriction	Recalculate NPSH
Bearing leads suction	Bearing or lubrication fault	MM-207 §6
Simultaneous, no consistent lag	Instrument or common-mode fault	Verify transmitters
Strainer DP leads suction by 0 to 10 min	Strainer fouling	Changeover per SOP

Table 2-1 — Lead/lag interpretation

3 NPSH Assessment

Available NPSH must be recalculated at the operating duty in force at the time of the events, not at design duty. Use the measured suction pressure, the fluid temperature at the suction flange and the actual static head.

1. Record suction pressure, suction temperature and flow at the time of the event.
2. Determine vapour pressure at the recorded suction temperature.
3. Compute available NPSH from static head, suction pressure and friction losses.
4. Read required NPSH from the manufacturer curve at the recorded flow.
5. Compute the margin. A margin below 0.5 m at any recorded operating point is a confirmed NPSH deficit.

> **NOTE** A margin that is adequate at design flow but deficient at the flows actually recorded is the most common finding in recurring-alarm investigations. Always assess at recorded duty.

4 Common Suction-Side Causes

- Strainer fouling, indicated by rising differential pressure preceding the event.
- Deaerator level control instability, indicated by level excursions toward the low-low setpoint.
- Partially closed or drifting suction valve, indicated by position feedback disagreeing with commanded position.
- Elevated feedwater temperature reducing available NPSH margin.
- Air ingress at flanges upstream of the pump, indicated by erratic suction pressure with no corresponding strainer trend.